MIL-R-22679 (AS)
Amendment 4
12 August 1968

Superseding
Amendment 3
28 September 1967

MILITARY SPECIFICATION
RECEIVER-TRANSMITTER RADIO RT-559/ASQ-58 and RT-559A/ASQ-58

This amendment forms a part of Military Specifications MIL-R-22679 (WEP) dated 1 November 1960 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-R-22679 (WEP) is changed to MIL-R-22679 (AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

Page 1, Title: Delete and substitute the title of this amendment.

Page 1, paragraph 1.1: Add

"NOTE: The RT-559A/ASQ-58 is compatible with Juliet-28 equipment."

Page 1, paragraph 1.2: Add

"(11) Juliet-28 (RT-559A/ASQ-58 only)
(12) C-8057/ARC (RT-559A/ASQ-58 only)
(13) C-2791A (RT-559A/ASQ-58 only)"

Page 2, paragraph 2.1, after "invitation for bids": Add

"except the dated document indicated below,"

Page 2, paragraph 2.1, under SPECIFICATIONS Military: Add

"MIL-I-6181B (29 May 1953) Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment (See 2.2(2))"

Page 2, paragraph 2.2: Add

"(1) Specification MIL-I-6181B is available from the Naval Air Systems Command, Washington 25, D. C. Attention: Avionics Division."

Page 2, Add:

"3.2.8 Antenna Conducted Interference - Antenna conducted interference shall meet the requirements of Specification MIL-I-6181B except that at not more than 9 of the 18 injection frequencies the antenna conducted interference due to the main receiver injection frequency shall not exceed 300 microvolts."

FSC 5821

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*Page 5, paragraph 3.3.2: Add

"The sensitivity of the auxiliary audio and the auxiliary guard audio shall be such that when an RF input of 5 microvolts (open circuit) modulated 30% at 1000 Hz is applied, the signal plus noise-to-noise ratio shall be not less than 3.0 db for the auxiliary audio and not less than 6 db for the auxiliary guard audio."

Page 5, paragraph 3.3.2.1: Delete and substitute

"3.3.2.1 Degraded or Unusable Channels Due to Internal Interference - Certain main receiver channel frequencies may be degraded or unusable due to internally generated interference. When the equipment is supplied with the guard frequency set for 243.0 mc, the main receiver sensitivity measurements shall be made with an input of 20 microvolts at frequencies of 241.1, 241.2, and 275.6 mc. Six unusable channels 242.1, 246.6, 253.3, 287.1, 322.8 and 339.0 mc are permitted which are independent of the guard receiver frequency."

Page 5, paragraph 3.3.4: Delete table and substitute:

| "Attenuation Below Resonant Response | TOTAL BANDWIDTH | | |
|---|---|---|---|
| | Guard Receiver | RT-559 Main Receiver | RT-559A Main Receiver |
| 6 db | 60 Kc min | 60 Kc min | 80 Kc min |
| 60 db | 400 Kc max | 200 Kc max | 200 Kc max" |

*Page 6, paragraph 3.3.9.1: Add

"The auxiliary guard output shall be 0.15 - 0.45 volts rms when loaded with 600 ohms."

Page 6, paragraph 3.3.9.3: Delete and substitute

"3.3.9.3 Auxiliary Circuit - The auxiliary circuits shall deliver only signals detected by the variable frequency receiver. When a 1000 - microvolt signal modulated 30% at 1000 cps is applied, the output shall be at least 0.25 volt, and when the modulation frequency is varied from 70 to 7000 cps, the output for any frequency in the range of 70 to 200 cps shall be within +4.5 and -3.0 db and in the range 200 to 7000 cps shall be within + 2 and -3 db of the output obtained at 1000 cps. The RT-559A/ASQ-58 shall not vary more than + 4 and - 4 db from the output obtained at 1000 cps when the modulation frequency is varied from 70 to 20,000 cps."

Page 7, paragraph 3.3.12.1.1: Delete table and substitute

| "Frequency Cps | Relative Output in db | |
|---|---|---|
| | RT-559/ASQ-58 | RT-559A/ASQ-58 |
| 150 | -8 or lower | -5 or lower |
| 300 | -3 to -7 | +4 to -4 |
| 600 | +1 to -2 | +3 to -3 |
| 1,000 | 0 (ref. level) | 0 (ref. level) |
| 3,000 | +2 to -2 | +2 to -2 |
| 6,000 | +2 to -2 | +3 to -3 |
| 10,000 | +1 to -5 | +4 to -4 |
| 15,000 | -- | +4 to -4 |
| 20,000 | -- | +4 to -4 |



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Page 8, paragraph 3.3.12.2: Delete and substitute

"3.3.12.2 Tone Modulation - Not less than 70 percent modulation of the r-f carrier shall occur when tone transmission is selected and the transmitter is keyed. The frequency of modulation shall be 1000 cps ± 20%."

Page 8, paragraph 3.3.13: Delete and substitute

"3.3.13 Spurious Radiation - The magnitude of all radiated signals on other than the carrier frequency shall be 40 db or more below the carrier signal except that harmonic frequencies shall be attenuated not less than 55 db at 450 mc with a linear increase to 75 db at 525 mc. The attenuation above 525 mc shall be 75 db or greater."

*Page 8, paragraph 3.3.16: Add

"Under the above condition the auxiliary audio and the auxiliary guard shall have a signal plus noise to noise ratio of not less than 20 db."

Page 10, paragraph 3.4.1.2: Delete "10-37/64": and substitute "10-11/16"

Page 10, paragraph 3.4.1.2: Add

"The front panel shall extend an additional 1-1/16 inches below the base plate and the pressure relief valve shall extend an additional 9/32 inches beyond the front of the case."

Page 12, paragraph 3.4.1.7.3: Add

"The auxiliary guard output of the RT-559A/ASQ-58 shall be terminated in a 600 ohm load, including means for measuring the audio output voltage."

Page 12, paragraph 3.4.1.7, under "Carbon Microphone Input" and under "Dynamic Mike Input":

Delete "Frequency range 150-10,000 cps" and substitute
"Frequency range 150-10,000 cps (RT-559/ASQ-58)
150-20,000 cps (RT-559A/ASQ-58)"

Page 12, paragraph 3.4.1.7.4: Add

"NOTE: 1 Vrms must be supplied when the RT-559A/ASQ-58 is used with Juliet equipment."

Page 13, after paragraph 5.1: Add

"5.2 Final Adjustments - Unless otherwise stated by the contract, the receiver-transmitter shall be shipped properly adjusted for signal-to-noise squelch and carbon microphone operation."

*An asterisk preceding a change denotes a change to be made to the specification which did not appear in Amendment 3.



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